



SPHERE FOUNDATIONS

PURPOSE

This practice establishes guidelines and recommendations for the design of foundations for spherical vessels. This practice covers calculation of loads and design of each of the components of the foundation.

SCOPE

This practice includes the following major headings:

- GENERAL
- VERTICAL LOADS
- HORIZONTAL LOADS
- DISTRIBUTION OF LOADS
- ESTIMATING PIER AND FOOTING DIMENSIONS
- FOOTING THICKNESS
- FOUNDATION DESIGN
- REFERENCES
- ATTACHMENTS

APPLICATION

This practice should be used for the design of ring type foundations for large spheroid vessels. Refer to Structural Engineering Practice 000.215.1224: Heater Foundations, for procedures for the design of smaller spheroid vessels on octagon mats.

GENERAL

The design of foundations for spheroid vessels presents a complex problem; therefore, some simplifying assumptions must be made in order to produce an adequate design within a reasonable time. Typically, the foundations for small spheres will be octagonal in shape, while larger spheres will be an annular or ring type foundation due to economics. Individual footings also may be used based on number and spacing of vessel columns.

VERTICAL LOADS

Definitions

Empty Weight: Fabricated weight of vessel, plus the weight of internals, piping, insulation, and platforms, generally taken from the vessel drawing.

Operating Weight: Empty weight plus weight of operating liquid or contents; generally taken from the vessel drawing.



SPHERE FOUNDATIONS

Test Weight: Empty weight plus the weight of water required for hydrostatic test; generally taken from the vessel drawing.

It should be determined if hydrostatic testing will actually be done in the field. Generally, it is desired to design for test as unforeseen circumstances may occur.

The above loads will be considered as Dead Loads when applying load factors.

HORIZONTAL LOADS

Wind

Wind loads will be calculated by the Structural Engineer using vessel and platform drawings and the following criteria:

- Spherical surface wind pressure on the projected area of the insulated vessel applied at the area centroid.
- Flat surface wind pressure on each platform horizontal area times a 0.5 factor applied at the height of each platform.

Wind pressure will be varied with height in accordance with the job specifications. Wind load calculations shall conform with Structural Engineering Practice 000.215.1215: Wind Load Calculations, for wind design considerations and procedures.

Wind loads calculated by hand methods will be compared to supplier provided load data. If the results compare favorably, the higher value will be used for foundation design. If the results do not compare favorably, resolve the discrepancy before proceeding.

Seismic

In earthquake zones, seismic forces as determined in accordance with job specifications and Structural Engineering Practice 000.215.1216: Earthquake Engineering, for seismic design considerations and procedures.

DISTRIBUTION OF LOADS

Vertical loads from dead weight are assumed equally distributed to each outer column.

Vertical loads due to overturning moments are assumed to be distributed to each column in proportion to the column distance from the neutral axis of the column group.



SPHERE FOUNDATIONS

Horizontal loads (wind or seismic) will be distributed to the spheroid vessel columns by tension bracing spanning between the columns. Both wind and seismic conditions shall be considered. For purposes of analysis, the sphere is assumed a rigid body with the applied horizontal external load being distributed to each braced panel in proportion to the stiffness of the panel in the direction of the applied load. The sum of the component panel shears parallel with the loading must be equal to the applied external load.

Panel Shear (V_p), the shear between columns which is transferred to the pedestals by the tension bracing, can be calculated as follows:

$$V_p = \frac{2V}{n} \cos \alpha$$

where

V = total horizontal force acting on the vessel and columns

n = number of columns

α = angle between panel and direction of horizontal force

For derivation of the above formula, refer to Attachment 01.

Load Combinations

The following load combinations will be used in the foundation design:

- Empty weight plus wind (or seismic)
- Operating weight plus wind (or seismic)
- Test weight plus partial wind , where partial wind is taken as 75% of basic wind speed.

Load Combinations shall be calculated in accordance with Structural Engineering Specification 000.215.00910: Structural Engineering Criteria.

ESTIMATING PIER AND FOOTING DIMENSIONS

It may be necessary to assume a trial size for the purpose of analysis. Pier sizes are determined from column base plate dimensions and minimum edge distances for anchor bolts. Footing sizes are estimated from the pier circle plus the width of the pier plus the desired edge distance.



SPHERE FOUNDATIONS

FOOTING THICKNESS

The footing thickness should be adequate to ensure sufficient rigidity for distributing soil bearing pressures in a uniform manner. Preliminary footing thickness for a ring type foundation may be determined from the following criteria, using the largest of the conditions:

$$t = 1'-0''$$

or

$$t = \frac{(x - b)}{3}$$

where:

t = Footing thickness, feet

x = Maximum projected distance between adjacent pier centerlines, feet

b = Width of concrete pier parallel with X , feet

Refer to the Reference section for various other equations by Bowles or Terzaghi, which may be used to check the rigidity of a footing. Many of these equations are based on the subgrade reaction of the soil. In all cases, engineering judgment should be used in interpreting the results from the above equations.

FOUNDATION DESIGN

Either a ring foundation or individual footings may be used depending on the number of supporting columns and column spacing. Individual footings are typically analyzed by hand. Ring type foundations may be analyzed by hand using simplified assumptions or by finite analysis methods. Several approaches for analysis of ring foundations using finite element analysis may be found in Foundation Analysis and Design or Analytical and Computer Methods in Foundation Engineering by Bowles. Finite element analysis may be performed using computer programs designated as approved reference systems.

Piers will be designed in accordance with recommended column design procedures outlined in ACI 318-05. Shear lugs may be utilized at column bases to resist high shear forces, which in turn may allow the use of smaller anchor bolts and column piers.

Anchor Bolts

Although the anchor bolts are usually specified by the manufacturer, they must be checked against the project specification. For design procedures for sizing anchor bolts including embedment lengths, refer to PIP Guidelines STE05121: Anchor Bolt Design Guide.



SPHERE FOUNDATIONS

Stability Ratio

The stability ratio (ratio of dead load resisting moment to overturning moment) will be according to job specifications, with a minimum of 1.5 for wind load and 1.0 for seismicload. If the dead load factor is 0.6 in accordance with ASCE 7-05, section 2 then the minimum factor of safety against overturning and sliding shall be 1.0. When calculating the stability ratio for seismic loads, 90 percent of the dead load should be used in accordance with Structural Engineering Practice: 000.215.1216:Earthquake Engineering.

Soil Bearing

If individual foundations are used, soil bearing pressure will be checked for maximum allowable on the diagonal. Soil bearing pressure used for reinforcing design will be computed on the flat.

Soil bearing pressure for ring type foundations will be checked for maximum allowable at outer edge of ring opposite applied lateral load.

Footing Shear And Moment

For ring type foundations supported on soil, the procedure described here is used to compute the shear force and moment values. For solid mats, refer to Structural Engineering Practice 000.215.1224: Heater Foundations, for procedures.

Find the maximum soil bearing value for the entire footing, usually located at the footing edge opposite the applied lateral force. For test condition, the soil bearing is assumed equally distributed under the entire footing.

The ring foundation may be analyzed by hand as a continuous beam spanning between piers. Check shear and moments at the critical cross sections adjacent to the piers and at midspan between the piers.

Beam shear and peripheral shear are checked to determine whether the footing thickness is adequate. Maximum value of the shearing force from all the loading cases is used to check the beam shear. It is required that the stresses on the critical periphery surrounding the pier be investigated as stated in ACI 318-05, Section 11.12.2 and 11.12.6.

Torsion due to twisting of the annular footing should be checked using procedures outlined in ACI 318-05, Section 11.6. For large torsional moments, the combining of radial and torsional steel in the form of closed ties (U shaped bars lapped at the top and bottom) should be considered.



SPHERE FOUNDATIONS

Footing Reinforcing

The minimum footing reinforcing will be #5 at 12 inches c/c top and bottom.

To minimize tagging of rebar and simplify rebar placement, longitudinally curved reinforcing (annular bars) should be divided in groups of the same radius, as opposed to bending each annular bar to a unique radius. Typically, the rebar fabricator and contractor installing the rebar can arrive at a satisfactory approach to this type of installation.

REFERENCES

ACI (American Concrete Institute) 318-05; Sections 11.12.2, 11.12.6, and Section 11.6

Bowles, Joseph E., Analytical and Computer Methods in Foundation Engineering, McGraw-Hill, 1974.

Bowles, Joseph E., Foundation Analysis and Design, McGraw-Hill, 1996.

Terzaghi, K, Evaluation of Coefficient of Subgrade Reaction, Geotechnique, London, 1955.

PIP 000.215.STC01015: Structural Design Criteria

PIP STE 05121: Anchor Bolt Design Criteria

Structural Engineering

Practice 000.215.1215: Wind Load Calculations

Structural Engineering

Practice 000.215.1216: Earthquake Engineering

Structural Engineering

Practice 000.215.1224: Heater Foundations

ATTACHMENTS

Attachment 01: (13Mar00)

Panel Shear (V_p) Equations

Attachment 02: (13Mar00)

Sample Design 1 - Sphere Foundation On Soil



SPHERE FOUNDATIONS



SPHERE FOUNDATIONS

Panel Shear (Vp) Equations

The shear stiffness of each panel is considered equal and the shear in each panel is:

$$V_i = K\Delta_i$$

Where: K = Stiffness

Δ_i = Displacement parallel with panel

The vessel is assumed a rigid body therefore the displacement in the direction of the external load must be equal for all braced panels.

Therefore:

$$\Delta_i = \Delta \cos \alpha_i$$

α_i = Angle between direction of external load and braced panel i

$$V_i = K\Delta \cos \alpha_i \quad \text{or} \quad K\Delta = \frac{V_i}{\cos \alpha_i}$$

The sum of the component panel shears parallel with the loading equals the applied external load, V

$$\sum_{i=1}^n V_i \cos \alpha_i = V \quad \text{substituting for } V_i, \quad \sum_{i=1}^n (K\Delta \cos \alpha_i) \cos \alpha_i = V \quad \text{then,}$$

$$\sum_{i=1}^n \cos^2 \alpha_i = \frac{V}{K\Delta}$$

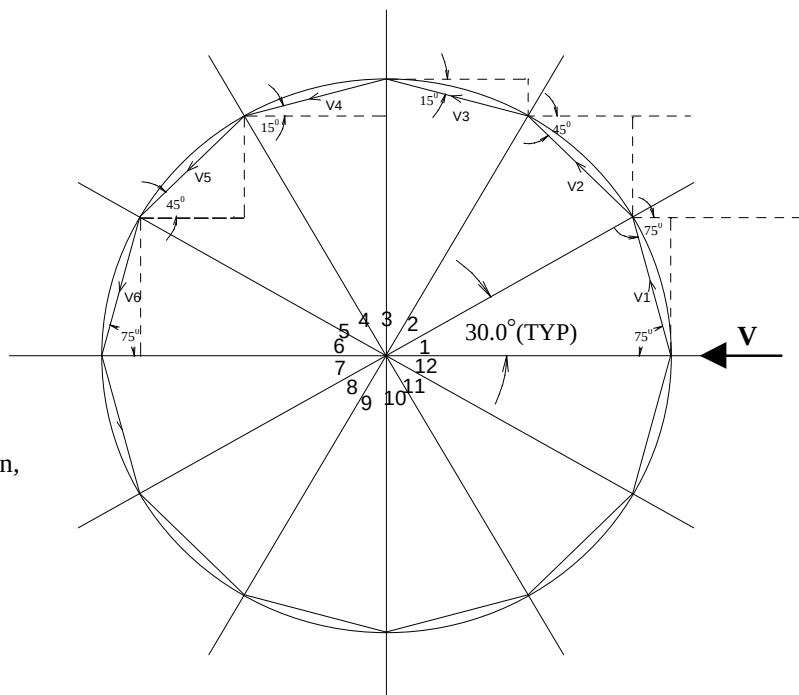
Substitute $K\Delta = \frac{V_i}{\cos \alpha_i}$ and

rearrange terms to get:

$$V_i = \frac{V \cos \alpha_i}{\sum_{i=1}^n \cos^2 \alpha_i}$$

For this geographical configuration, mathematically:

$$\sum_{i=1}^n \cos^2 \alpha_i = \frac{n}{2}$$





SPHERE FOUNDATIONS

Panel Shear (Vp) Equations

Therefore:

$$V_i = \frac{2V}{n} \cos \alpha_i$$

Where: V_i = Panel shear in brace bay i

V = Total applied external load

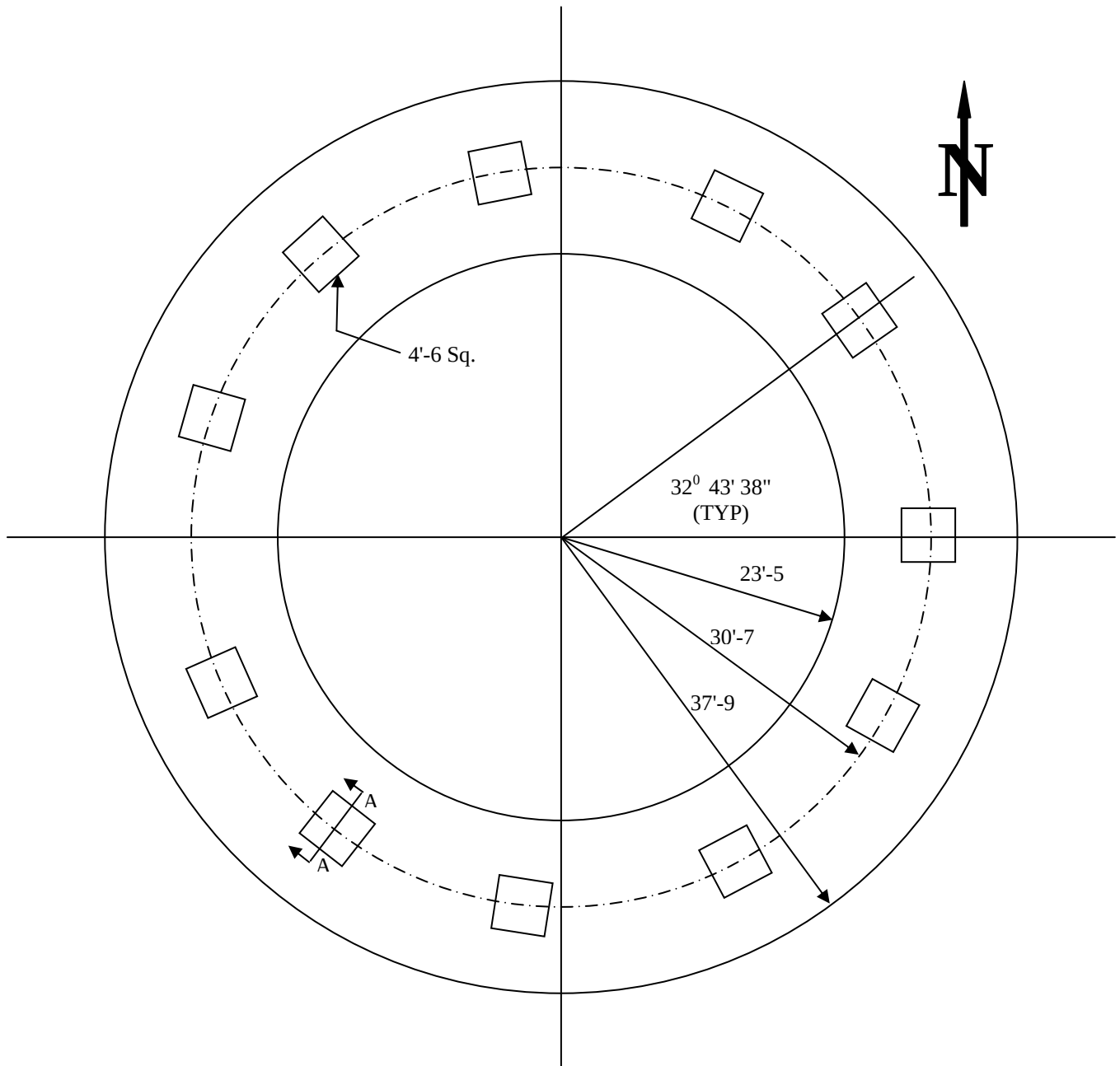
n = Number of sides

α_i = Angle between direction of external load and braced panel i

For the 12 panel system with $n = 12$

$$V_1 = \frac{V}{6} \cos 75^\circ \quad V_2 = \frac{V}{6} \cos 45^\circ \quad V_3 = \frac{V}{6} \cos 15^\circ$$

SPHERE FOUNDATIONS



FOUNDATION PLAN

SPHERE FOUNDATIONS

Reference:

ACI 318-05

IBC 2006

ASCE 7-05

Design Data

$f'_c = 3000$ psi
 $f_y = 60,000$ psi

Soil Wt = 120 PCF
Soil Bearing = 4.0 KSF (NET)
(No increase allowed for wind)

Vessel Wts:

Fab. Wt. = 1239.7 K
Oper. Wt. = 7977.2 K
Test Wt. = 10,000 K (Including column fireproofing)
Spec. gravity of the product = 0.51

Avg. wall thickness of the sphere = $t = 1.91$ "
Sphere supported on 11 pipe columns equally spaced
(34" O.D., 11" / 32 Wall Thk W/ 2" of fireproofing; Base Pl Thk = 1 1/2")

Wind Loads:

(Per job spec. and ASCE 7-05)

HEIGHT	0-30 Ft	30-50 Ft	50-100 Ft
Flat	20 PSF	25 PSF	30 PSF
Curved**	12 PSF	15 PSF	18 PSF

*Do not consider seismic loads in design of this particular sphere foundation (i.e. seismic zone = 0).

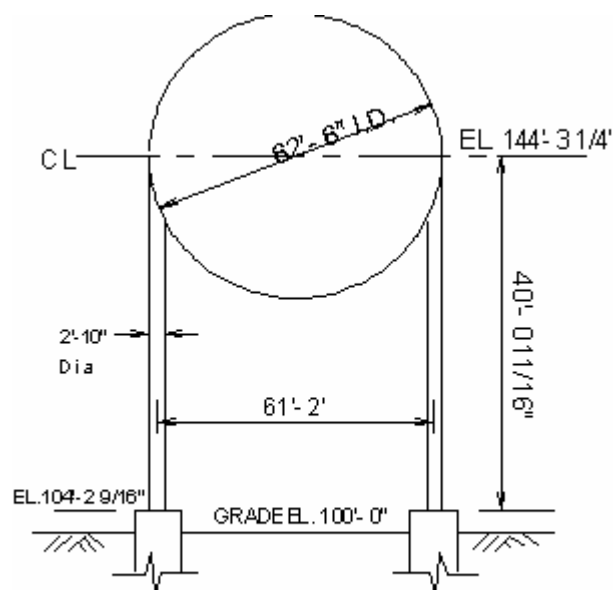
** 0.6 x Wind load on Flat.

Wt Of Fireproofing On The Columns: (USE H = 40 Ft - Conservative)

$$\text{Conc. F.P. Volume} = 40 \frac{\pi}{4} (3.17^2 - 2.83^2) = 64.1 \text{ FT}^3$$

$$\text{Conc. F.P. Wt./Col.} = 64.1 (0.150 \text{ KCF}) = 9.62 \text{ K/Col}$$

$$\text{Total F.P. Wt.} = 9.62 (11) = 105.82 \text{ K}$$





SPHERE FOUNDATIONS

Total Design Weights:

$$\text{Empty Wt.} = 1239.7 + 105.82 = 1345.52 \text{ K}$$

$$\text{Oper Wt.} = 7977.2 + 105.82 = 8083.02 \text{ K}$$

$$\text{Test Wt.} = 10000 + 105.82 = 10105.82 \text{ K}$$

Calculate Wind Loads:

$$\text{Basic Dia. of Vessel} = 62'-6" + 2 (1.91") = 62.82 \text{ Ft}$$

$$\text{Dia F.P. columns} = 34" + 2 (2 ") = 38" = 3.17 \text{ Ft}$$

Use max. wind pressure (50-100 Ft) for the entire vessel:
Curved surface wind = 18 PSF

For brevity wind load calculations are not shown. Refer to technical practice 000-215-1215 for procedures.

$$V_{\text{wind}} = 67.2 \text{ K}$$

$$V_{\text{wind, partial}} = (0.75^2) * 67.2 \text{ K} = 37.8 \text{ K}$$

Note: Partial wind is taken as 75% of Basic wind speed. Intensity is proportional to squarer of wind speed.

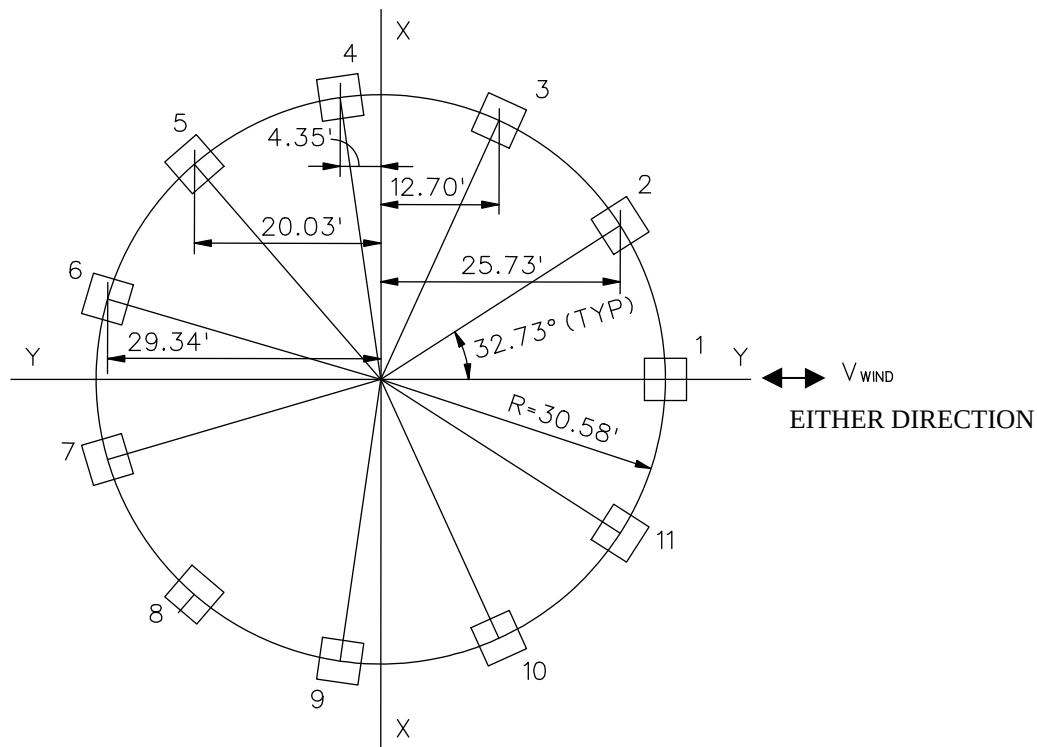
Determine Vertical Forces At Top Of Piers:

$$\text{Wind force} = 67.2 \text{ K}$$

$$\text{Moment} = 67.2 \text{ K} (40.06 \text{ Ft}) = 2692 \text{ K-Ft}$$

SPHERE FOUNDATIONS

Case 1: Wind Applied at Column 1



$$P = \frac{Mc}{I}$$

$$I_{xx} = \frac{Nd^2}{8} = \frac{11(61.167)^2}{8} = 5144$$

$$P_1 = 2692(30.58)/5144 = \pm 16.00K$$

$$P_{2,11} = 2692(25.73)/5144 = \pm 13.47K$$

$$P_{3,10} = 2692(12.70)/5144 = \pm 6.65K$$

$$P_{4,9} = 2692(4.35)/5144 = \pm 2.28K$$

$$P_{5,8} = 2692(20.03)/5144 = \pm 10.48$$

$$P_{6,7} = 2692(29.34) / 5144 = \pm 15.35K$$

Case II: Wind Applied Between Columns 1 and 11

$$P = \frac{Mc}{I}$$

$$I_{xx} = \frac{Nd^2}{8} = 5144$$

$$P_{1,11} = \pm 15.35K$$

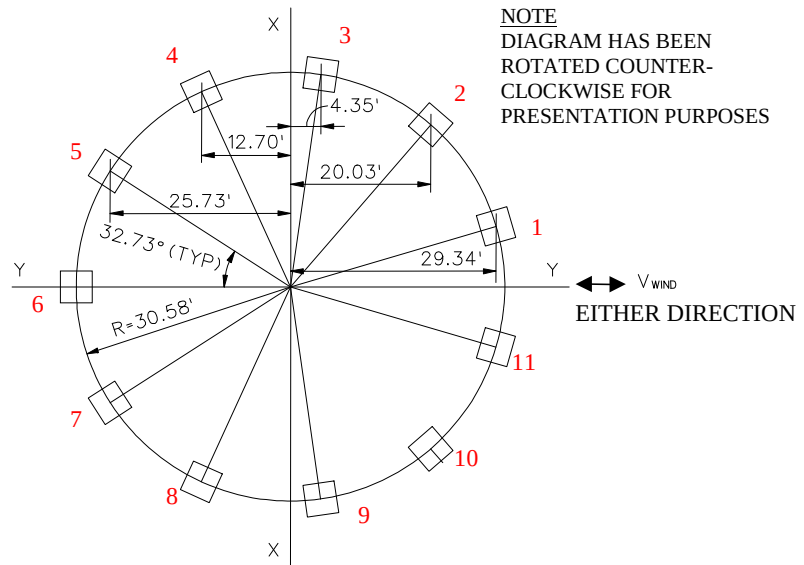
$$P_{2,10} = \pm 10.48K$$

$$P_{3,9} = \pm 2.28K$$

$$P_{4,8} = \pm 6.65K$$

$$P_{5,7} = \pm 13.47 K$$

$$P_6 = \pm 16.00K$$



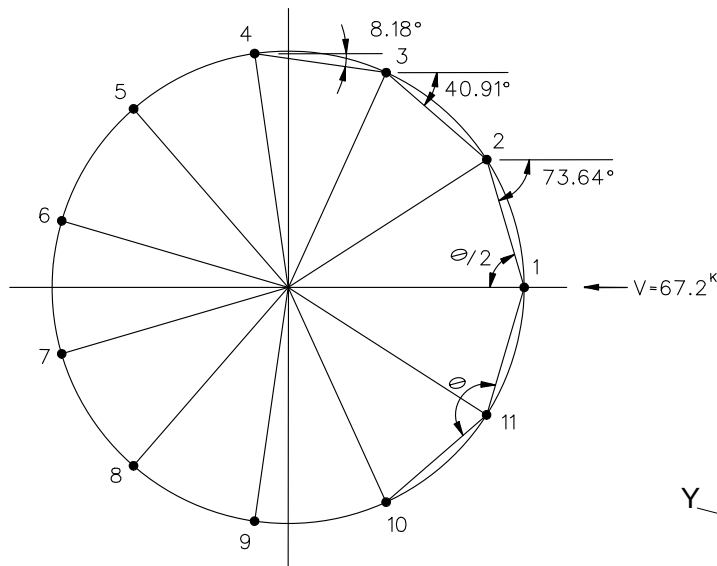
Determine The Shear In Each Panel:

Panel Shear: $V_p = \frac{2V}{N} \cos \alpha$ (α = Angle between panel and direction of lateral force)

$$V_{P_{\max}} = \frac{2V}{N} \text{ (Refer to Attachment No.1 for derivation of equation)}$$

SPHERE FOUNDATIONS

Case 1: Direction of Force Acting at Panel Intersection



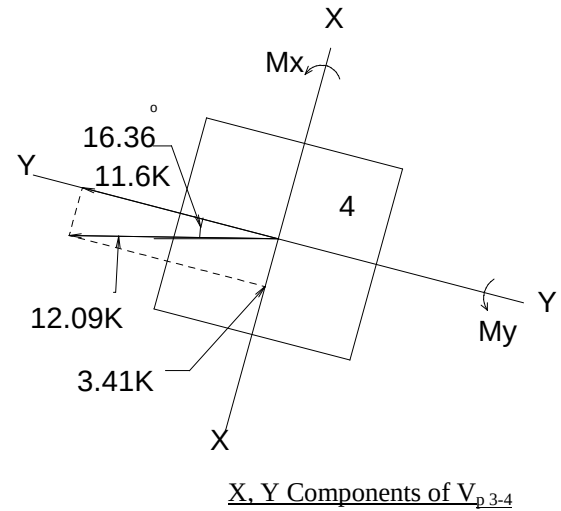
$$\theta = \frac{N-2}{N} 180^\circ$$

$$\frac{\theta}{2} = \frac{9}{2(11)} (180) = 73.64$$

$$V_{p11-2} = \frac{2(67.2)}{11} \cos 73.64^\circ = 3.44K$$

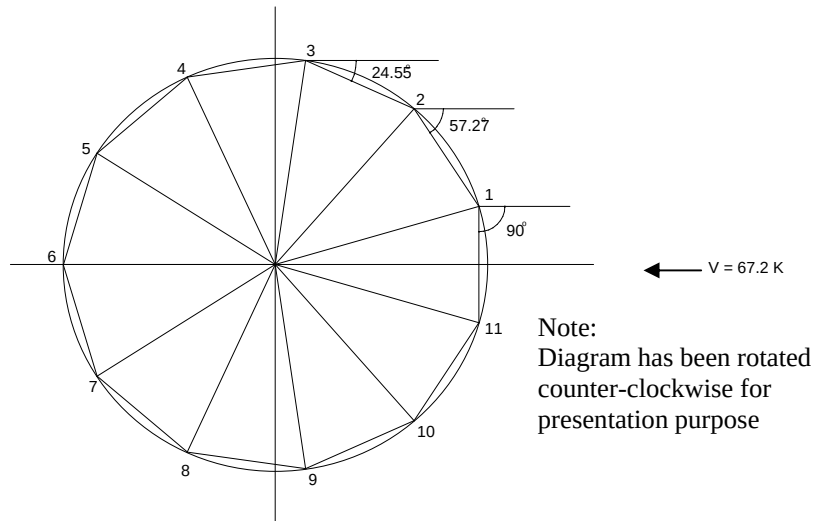
$$V_{p10-3} = \frac{2(67.2)}{11} \cos 40.91^\circ = 9.23K$$

$$V_{p9-4} = \frac{2(67.2)}{11} \cos 8.18^\circ = 12.09K$$



SPHERE FOUNDATIONS

Case II: Direction of Force Acting Normal To Panel



$$V_{p11-1} = \frac{2(67.2)}{11} \cos 90^\circ = 0.0$$

$$V_{p10-2} = \frac{2(67.2)}{11} \cos 57.27^\circ = 6.61K$$

$$V_{p9-3} = \frac{2(67.2)}{11} \cos 24.55^\circ = 11.11K$$

⇒ THIS CASE DOES NOT CONTROL

Load Conditions For Design

Test + Partial Wind (75% of Basic Wind Speed)
 Operating + Wind
 Empty + Wind

Test Load	=	10106 / 11	=	919 K / Col
Oper. Load	=	8083 / 11	=	735 K / Col
Empty Load	=	1346 / 11	=	122 K / Col

Anchor Bolt Check:

Tension: Bolts will not experience any tension



SPHERE FOUNDATIONS

Shear: Max. Shear = 12.09 K (Case I, Col. 3, Sht 7)

Friction Force $\approx 0.6(122K) = 73.2K$ where $0.6 = \mu$ conc/steel (Empty Case) .

Note that friction could be counted since Seismic is not applicable in this particular combination.

Bolts will not experience any shear because friction force will never be overcome.

USE: 2-3/4" Φ TYPE ASL BOLTS/PIER	-Refer to PIP guidelines STE05121 "Anchor Bolt Design Criteria" for design procedures.
--	--

Projection (from TOC) = N2 + 1 1/2" thk Base plate + 1" grout = 4.75", say 5"

Pier Design: (DESIGN AS A COLUMN)

Base Plate = 48" Sq. Bolt Spacing = 42" c/c

Try: 4'-6" X 4'-6" pier w/ 2'-6" thk. Ftg (Bot. of Ftg. El.=95'-0")

Height of Pier = $6' - 8\frac{9}{16}" \approx 6.75\text{ Ft}$

Pier Wt. = $(4.5)^2(0.150)(6.75) = 20.5K$

Load Condition I: TEST WEIGHT + PARTIAL WIND

(Case I: Col 4, Sht 7)

$M_x = 11.6 \times 6.75 \times 37.8/67.2 = 44.04 \text{ K-ft}$

$M_y = 3.41 \times 6.75 \times 37.8/67.2 = 12.95 \text{ K-ft}$

$P = 919 \text{ K} + 20.5 \text{ K} + 6.65 \times (37.8/67.2) = 943.2 \text{ K}$

$P_u = 1.2 (919 + 20.5) + 1.6 (6.75 \times 37.8/67.2) = 1133.4 \text{ K}$

$M_{u,x} = 1.6 M_x = 70.5 \text{ K-ft}$

$M_{u,y} = 1.6 M_y = 20.7 \text{ K-ft}$

Check Biaxial Capacity: (Use PCA COL to design the piers / columns)

Refer Sheets 18 & 19 for the PCA COL Results and the Interaction diagram, respectively.

Assume minimum reinforcement ($\rho = 0.01$) $A_{st} = 0.01(54)^2 = 29.2 \text{ in}^2$



SPHERE FOUNDATIONS

Load Condition II: OPER + WIND

(Case I: Col 4, Sht. 7)

$$M_x = 11.60K (6.75) = 78.3 \text{ K-Ft}$$

$$M_y = 3.41K (6.75) = 23.0 \text{ K-Ft}$$

$$P = 735K + 20.5K - 6.65K = 748.9 \text{ K}$$

$$P_u = 1.2 (735 + 20.5) - 1.6 (6.65) = 896 \text{ K}$$

$$M_{ux} = 1.6 (78.3) = 125.3 \text{ K-Ft}$$

$$M_{uy} = 1.6 (23.0) = 36.8 \text{ K-Ft}$$

Check Biaxial Capacity: (Use PCA COL to design the piers / columns)

Refer Sheets 18 & 19 for the PCA COL Results and the Interaction diagram, respectively.

NOTE: However, for pier design minimum reinforcement requirement is governing for both the combinations above.

$$\rho = 0.01$$

$$A_{s,req} = 0.01(54)^2 = 29.2in^2$$

Use : 24 #10^S (6 EA. face) w/ #4 Ties ($A_c = 30.48 \text{ in}^2$)

$$A_{s,prov} = 30.48in^2$$

Check Development (PER ACI 318-05, SECTION 12.5)

SPHERE FOUNDATIONS

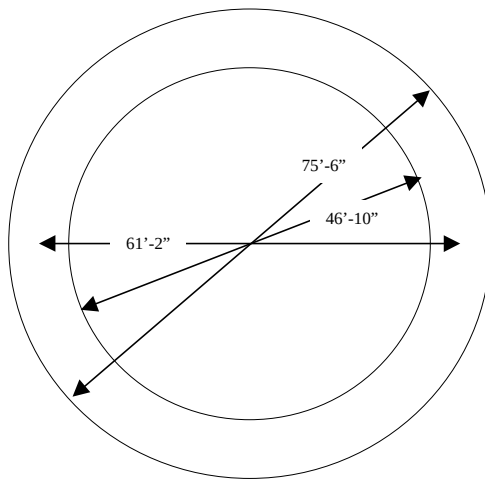
For #10 Bars: $\ell_{dh} = \left(\frac{0.02 \Psi_e \lambda f_y}{\sqrt{f'_c}} \right) d_b = 28"$ $A_{s,req} / A_{s,prov} = 0.95$

$(\ell_{dh})_{mod} = 28" (0.95)(0.7) = 18.6"$ 0.7 per ACI 318 – 05 Cl. 12.5.3 (a)

$T_{REQ'D} = 3" + 2(1") + 18.6" = 23.6" < 30"$ O.K.

Footing Design

Try Ring Type Foundation: O.D. = 75'-6" Width = 14'-4"
 I.D. = 46'-10" Thk = 2'-6"



$$\text{Area} = \frac{\pi}{4} (75.5^2 - 46.83^2) = 2755 \text{ Ft}^2$$

$$\text{Footing Wt.} = 2755(2.5)(0.15) = 1033 \text{ K}$$

$$\text{Soil Wt.} = [2755 - (4.5)^2(11)](2.5)(0.12) = 760 \text{ K}$$

$$\text{TOTAL} = 1793 \text{ K}$$

$$\begin{array}{llllll} \text{Empty Wt.} & = 1346 \text{ K} & + 11(20.5) \text{ K} & + 1793 \text{ K} & = 3364.5 \text{ K} \\ \text{(Sht 4)} & \text{(Col.)} & \text{(Pier)} & \text{(Ftg.)} & \end{array}$$

$$\text{Oper. Wt.} = 8083 \text{ K} + 11(20.5) \text{ K} + 1793 \text{ K} = 10101.5 \text{ K}$$

$$\text{Test Wt.} = 10106 \text{ K} + 11(20.5) \text{ K} + 1793 \text{ K} = 12124.5 \text{ K}$$

SPHERE FOUNDATIONS

$$\text{Moment} = 67.2 \text{ K} (6.75 + 2.5) = 621.6 \text{ K-ft}$$

$$\& \text{ For Partial Wind, Moment} = 621.6 \times 37.8 / 67.2 = 349.7 \text{ K-ft}$$

$$\text{Soil Bearing (Gross)} = 4.0 + 5 (0.12) = 4.6 \text{ KSF @ 5 Ft Below Grade}$$

$$I_{FTG} = \frac{\pi}{64} [(75.5)^4 - (46.83)^4] = 1358905 \text{ Ft}^4$$

1.) TEST + PARTIAL WIND :

$$SB = \frac{P}{A} \pm \frac{Mc}{I} = \frac{121245 \text{ K}}{2755 \text{ Ft}^2} \pm \frac{349.7 \text{ Ft}(37.75)}{1358905} = 4.41 \text{ KSF or } 4.39 \text{ KSF} < 4.6 \text{ KSF} \quad \underline{\text{O.K.}}$$

2.) OPERATING + WIND:

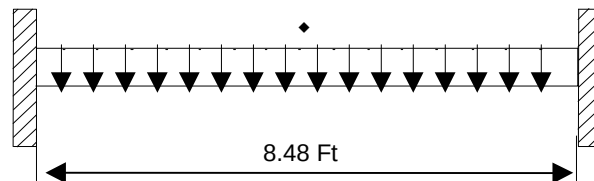
$$SB = \frac{P}{A} \pm \frac{Mc}{I} = \frac{101015 \text{ K}}{2755 \text{ Ft}^2} \pm \frac{621.6 \text{ Ft}(37.75)}{1358905} = 3.68 \text{ KSF or } 3.65 \text{ KSF} < 4.6 \text{ KSF} \quad \underline{\text{O.K.}}$$

Stability Ratio (S.R.) = O.K. by inspection

3.) EMPTY + WIND:

$$SB = \frac{P}{A} \pm \frac{Mc}{I} = \frac{3364.5}{2755} \pm \frac{621.6(37.75)}{1358905} = 1.24 \text{ KSF or } 1.20 \text{ KSF} < 4.6 \text{ KSF} \quad \underline{\text{O.K.}}$$

S.R. = O.K. by inspection



Foundation Size Adequate For Soil Bearing And Stability

One Way Shear:

(Note: Test Condition Controls. Since Partial wind effect is very small it is neglected and 1.4 test weight is considered)

Assume beam action between piers. One foot wide beam with both ends fixed.

$$\text{Chord length} = 2r \sin \frac{A}{2} = 2(30.58) \sin(32.73/2) = 17.23 \text{ Ft}$$

SPHERE FOUNDATIONS

Critical Shear occurs @ "d" from face of piers

$$d = 30" - 3" - 1 \frac{1}{2} (1") = 25.5"$$

$$\text{Effective beam length (L)} = 17.23 \text{ Ft} - 2(25.5/12) - 4.5 \text{ Ft} = 8.48 \text{ Ft}$$

$$S.B. = 1.4(4.40 \text{ KSF}) = 6.16 \text{ KSF}$$

$$w = 6.16 - 1.4[2.5(0.15) + 2.5(0.12)] = 5.22 \text{ KSF}$$

$$V_{u,MAX} = W L / 2 = 5.22(8.48) / 2 = 22.1 \text{ k}$$

$$v_u = \frac{22.1 \times 1000}{0.75(12)(25.5)} = 96.3 \text{ PSI} < 2\sqrt{f'_c} = 109.5 \text{ PSI} \quad \underline{\text{O.K.}}$$

Check Shear @ side-ways i.e in cross-section: ("d" from face)

$$4.92 - 25.5/12 = 2.80 \text{ Ft}$$

$$V_u = 2.80 (5.22) = 14.62 \text{ K}$$

$$v_u = \frac{14.62 \times 1000}{0.75(12)(25.5)} = 63.7 \text{ PSI} < 2\sqrt{f'_c} \quad \underline{\text{O.K.}}$$

Check Punching Shear:

$$d = 25.5"$$

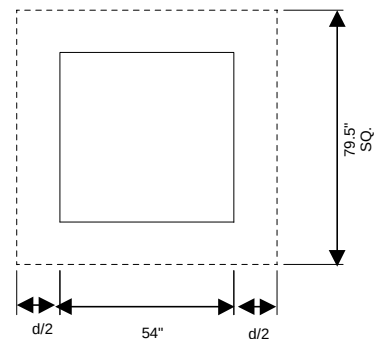
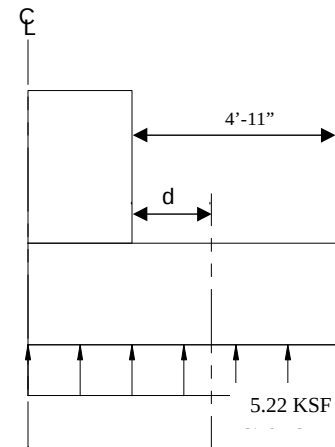
Test condition:

$$P_u = 1.4(919 + 20.5) = 1315.3 \text{ K}$$

$$b_o = 4(79.5") = 318"$$

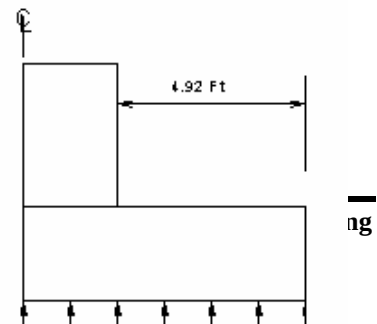
$$v_u = \frac{1315.3 \times 1000}{0.75(318)(25.5)} = 216.27 \text{ PSI} < 4\sqrt{f'_c} = 219.1 \text{ PSI} \quad \underline{\text{O.K.}}$$

∴ Use 2'-6" Thk Footing



Footing Steel:

Radial Reinforcing: (Test Condition)



SPHERE FOUNDATIONS

$$\begin{aligned}\text{Moment @ face of Ped} = M_u &= w l^2 / 2 = 5.22(4.92)^2 / 2 \\ &= 63.2 \text{ K-ft}\end{aligned}$$

$$F = \frac{b d^2}{12000} = \frac{12(25.5)^2}{12000} = 0.65$$

$$K_u = \frac{M_u}{F} = \frac{63.2}{0.65} = 97.23; a_u = 4.40$$

$$w = 5.22 \text{ KSF}$$

$$A_s = \frac{M_u}{a_u d} = \frac{63.2}{4.40(25.5)} = 0.56 \text{ in}^2 / \text{Ft}, \frac{4}{3} A_s = \underbrace{0.75 \text{ in}^2 / \text{Ft}}_{\text{CONTROLS}}$$

$$A_{s,\min} = 0.0033(12)(25.5) = 1.01 \text{ in}^2 / \text{Ft}$$

Use #8 @ 12" c/c ($A_s = 0.79 \text{ in}^2/\text{Ft}$) - Bottom, Radial

For Radial Top Steel use minimum temperature steel - O.K. by inspection
 (Ref.: ACI 318-05, Sec 7.12.1 and 7.12.2)

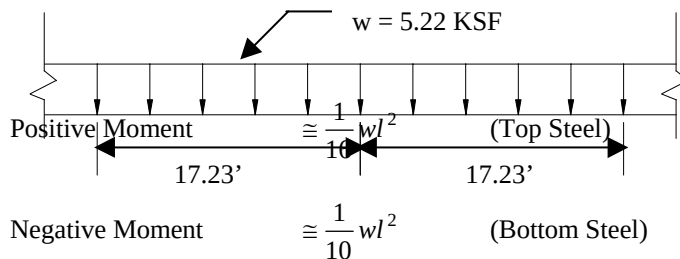
$$A_s = 0.0018 (12) (30) = 0.65 \text{ in}^2/\text{Ft}$$

Use #8s @ 12" c/c ($A_s = 0.79 \text{ in}^2/\text{Ft}$) - Top Radial

Longitudinal Steel: (Test Condition)

Assume footing acts as a continuous beam between piers.
 Take 1'-0" strip (Chord length = 17.23 Ft)

Assume: (Ref.: ACI 318-05, sec. 8.3)



SPHERE FOUNDATIONS

Check Torsional strength of footing: (Ref.: ACI 318-02, Sec. 11.6.1(a))

$$T_n = \phi \sqrt{f'_c} \left(\frac{A_{cp}^2}{p_{cp}} \right)$$

A_{cp} = Area enclosed by outside perimeter of concrete section, in²

= Depth of beam x width of beam = 30" x 172" = 5160 in²

p_{cp} = Outside perimeter of the concrete cross section, in

= 2(30"+172") = 404 in

$$T_n = 0.75 \sqrt{3000} \left(\frac{5160^2}{404} \right) / (12 \times 1000) = 225.6 \text{ Kft} > 56.51 \text{ Kft} \quad (\text{Moment per column} = 621.6 / 11)$$

Neglect torsional effects on ring footing

Bottom Steel:

$$M_u = \frac{5.22(17.23)^2}{10} = 155 \text{ Kft}$$

$$F = \frac{bd^2}{12000} = \frac{12(25.5)^2}{12000} = 0.65$$

$$K_u = \frac{M_u}{F} = \frac{155}{0.65} = 238.5; a_u = 4.25$$

$$A_s = \frac{M_u}{a_u d} = 153.2 / (4.25 \times 25.5) = \underbrace{1.41 \text{ in}^2 / \text{Ft}}_{\text{CONTROLS}} \quad \frac{4}{3} A_s = 1.88 \text{ in}^2 / \text{Ft}$$

$$A_{s,\min} = 0.0033(12)(25.5) = 1.01 \text{ in}^2 / \text{Ft}$$

Top

Use #10^S @ 10" c/c ($A_s = 1.52 \text{ in}^2 / \text{Ft}$)- Bottom Longitudinal (Splice Length = 8'-0")

Steel:

d=30" – 2" – 1
½ (1") = 26.5"

$$M_u = \frac{5.22(17.23)^2}{16} = 96.85 \text{ Kft}$$



SPHERE FOUNDATIONS

$$F = \frac{bd^2}{12000} = \frac{12(26.5)^2}{12000} = 0.70$$

$$K_u = \frac{M_u}{F} = \frac{96.85}{0.70} = 138.4; \quad a_u = 4.36$$

$$A_s = \frac{M_u}{a_u d} = \frac{96.85}{4.36(26.5)} = 0.84 \text{ in}^2 / \text{Ft} \Rightarrow \text{CONTROLS}$$

$$\frac{4}{3} A_s = 1.12 \text{ in}^2 / \text{Ft}$$

$$A_{s \min} = 0.0033(12)(26.5) = 1.05 \text{ in}^2 / \text{Ft}$$

Use #8^s@9" c/c ($A_s=1.05 \text{ in}^2 / \text{FT}$) - Top Longitudinal (Splice Length = 5'-0")

**NOTE : FOLLOWING COMPUTER OUTPUT IS NOT UPDATED PER ACI 318 – 05 . IT SHALL BE
UPDATED ONCE THE UPDATED SOFTWARE COMPLYING ACI318 - 05 IS RELEASED**



SPHERE FOUNDATIONS

SP:SP:00 Licensed to: Fluor Corporation, All Cities, All Over

SPHERE

General Information:

File Name: H:\ARCHIVE\215\UPDATE1\15\000_215\2\UPDATED\SPHERE.CML
Project: FLUOR PRACTICE 000 215 1223
Column: CML 8 Engineer: HANSEN
Code: ACI 308-95 Rebar: English
Run Option: Design Slenderness: Not considered
Run Axis: Global Column Type: Structural

Material Properties:

$f'_c = 3 \text{ ksi}$ $f_y = 60 \text{ ksi}$
 $E_c = 3122.02 \text{ ksi}$ $E_s = 29000 \text{ ksi}$
 $f_c = 2.55 \text{ ksi}$ Rupture strain = Infinity
Ultimate strain = 0.003 in/in
Beta1 = 0.85

Section:

Rectangular: Width = 54 in Depth = 54 in
Gross section area, $A_g = 2916 \text{ in}^2$
 $I_x = 708588 \text{ in}^4$ $I_y = 708588 \text{ in}^4$
 $S_x = 8 \text{ in}$ $S_y = 8 \text{ in}$

Reinforcement:

Rebar Database: ASTM A615

Size	Diam (in)	Area (in ²)	Size	Diam (in)	Area (in ²)	Size	Diam (in)	Area (in ²)
# 3	0.38	0.11	# 4	0.50	0.32	# 5	0.63	0.31
# 6	0.75	0.44	# 7	0.88	0.68	# 8	1.00	0.79
# 9	1.13	1.00	# 10	1.27	1.27	# 11	1.41	1.56
# 14	1.68	2.25	# 18	2.26	4.00			

Confinement: User-defined; #4 ties with #10 bars. #4 with larger bars.
 $\phi(s) = 0.85$, $\phi(h) = 0.9$, $\phi(c) = 0.65$

Layout: Rectangular
Pattern: All Sides Equal (Cover to transverse reinforcement)
Total steel area, $A_s = 30.40 \text{ in}^2$ at 1.05%
24 #10 Cover = 2 in

Factored Loads and Moments with Corresponding Capacities: (see user's manual for notation)

No.	P_u kip	M_{ux} k-ft	M_{uy} k-ft	P_{uax} k-ft	P_{uay} k-ft	P_{uax}/P_u
1	884.0	125.3	34.8	3236.7	975.1	36.620



SPHERE FOUNDATIONS

